Course Project for SE

Full Project



Group Name: _____

Version: _____

Instructor Name:

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D2 – Requirements Elicitation

Project: Digital multiplayer board-race game (Fall 2026)

Team: Group of 3

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1. Scope statement

This document outlines the user requirements for a digital multiplayer board-race game, the game component our team is building for the Software Engineering and Project Management course in Fall 2026. The requirements were elicited from the client during Sprint 1 and are the input for the system requirements in D3.

The client is the teaching assistant team of the course, acting as the product client. They want a desktop or web game for 2 to 4 players, inspired by traditional race games such as Ludo [1], in which each player moves several pieces from the starting area to the final destination before the others. The game must support Human vs Human and Human vs Computer play, enforce valid moves, track turns and progress automatically and identify the winner.

The client has asked us to clarify the detailed rules in a meeting before the final SRS. Our current working choice is Ludo-style rules. Stories that depend on specific rules (REQ-PLAYER10 and REQ-PLAYER11) are preliminary and will be updated after that meeting.

2. User classes

Hassan, 19, Casual Player. Hassan is a second-year student who plays board games with friends during breaks. He knows the basic rules but forgets details, such as when a piece may leave the start. He never reads instructions, wants to start a game in a few clicks and always see whose turn it is. When his friends are busy, he plays against the computer.

Mariam, 22, Experienced Player. Mariam plays race board games often, usually with three friends at once, and knows the rules well. She gets annoyed by rule mistakes and slow games. She wants fair random dice, correct captures, a clear view of everyone's progress and a clear announcement of the winner.

3. User requirements

User stories follow the format: *As a [role], I want [goal], so that [benefit].*

Game setup

- **REQ-PLAYER1.** As a player, I want to start a new game from the main menu, so that I can begin playing quickly.
- **REQ-PLAYER2.** As a player, I want to choose between 2, 3 or 4 players, so that the game fits the people I have.
- **REQ-PLAYER3.** As a player, I want to set each seat as human or computer, so that I can play Human vs Human or Human vs Computer.

- **REQ-PLAYER4.** As a player, I want to enter a name for each human player, so that it is clear who is who during the game.
- **REQ-PLAYER5.** As a player, I want to read a short summary of the rules from the menu, so that I can learn or check them easily.

Gameplay

- **REQ-PLAYER6.** As a player, I want to roll a virtual dice on my turn, so that movement is random and fair.
- **REQ-PLAYER7.** As a player, I want my piece to move automatically when only one move is possible, and to choose the piece when there are several, so that I can plan my strategy without counting squares.
- **REQ-PLAYER8.** As a player, I want to see whose turn it is at all times, so that I never get confused.
- **REQ-PLAYER9.** As a player, I want the game to allow only valid moves and show a clear message when I try an invalid one, so that I can correct it and the game stays fair.
- **REQ-PLAYER10.** As a player, I want an opponent's piece to return to its start when I land on it, so that I can compete with my opponents.
- **REQ-PLAYER11.** As a player, I want a six to let me bring a piece out of the start and roll again, so that the game follows the standard rules.
- **REQ-PLAYER12.** As a player, I want to see each player's progress (pieces at start, on the board and finished), so that I know who is ahead.
- **REQ-PLAYER13.** As a player, I want the computer opponent to play its turn automatically and quickly, so that the game does not slow down.
- **REQ-PLAYER14.** As a player, I want the board, pieces and dice to update immediately after every action, so that the game feels responsive and easy to follow.

Game end

- **REQ-PLAYER15.** As a player, I want the winner to be identified and announced automatically when all of one player's pieces reach the final destination, so that the result is clear.
- **REQ-PLAYER16.** As a player, I want to restart the current game at any point, so that I can start over after a mistake.
- **REQ-PLAYER17.** As a player, I want to start a new game after one ends, so that we can have a rematch without going back through the menu.
- **REQ-PLAYER18.** As a player, I want to quit the game at any point, so that I can leave when I need to.

The client also expects the game not to crash during normal play and to allow future features and rule changes. These are system-level constraints and will be specified in D3.

Changelog

Version	Date	Change
v1	2026-10-18	Initial version
v2	2026-10-18	Aligned with the client's initial requirements: 2 to 4 players, Human vs Computer mode, invalid-move handling, player progress, responsiveness. Stories renumbered (now 18).

References

[1] "Ludo (board game)." Wikipedia: The Free Encyclopedia. Wikimedia Foundation, Inc. https://en.wikipedia.org/wiki/Ludo_(board_game)